Grouped Attention-Based Spectral–Spatial Feature Learning for Collaborative Hyperspectral and LiDAR Classification

Yan Huang, Binghua Lu, Zhizhong Zheng, Kang Ni, *Member, IEEE*, and Liang Xiao, *Senior Member, IEEE*

***Abstract*—The attention mechanism, through its dynamic weight allocation and feature enhancement, effectively improves the representation capability of land-cover features in hyperspectral imaging and light detection and ranging (LiDAR) data. However, most implementations of attention mechanisms tend to neglect the local attributes of land-cover features and the prior information inherent in hyperspectral imagery (HSI) data. Grouped feature learning mechanisms can focus on different characteristics or structures of land-cover objects. Based on this, this article integrates a grouping mechanism into the convolutional neural networks (CNNs)-Transformer framework and proposes a grouped attention-based spectral–spatial feature learning (GAS^2-FL) network. Exploiting the low-rank prior of HSI data, the network incorporates a spectral–spatial feature learning module and embeds a grouped feature learning strategy to model spectral information interactions among heterogeneous features. Furthermore, a grouped token mechanism and a nonlocal token learning mechanism are developed to enhance the interaction of local attribute features and the learning of fused representations. The superior performance of the proposed network on four HSI and LiDAR classification benchmarks comprehensively demonstrates that its grouped feature learning mechanism is highly effective at extracting and representing the discriminative features essential for land-cover characterization. The source codes will be available at https://github.com/RSIP-NJUPT/GAS2-FL**

***Index Terms*—Classification, feature enhancement, grouped mechanism, hyperspectral image, transformer.**

Received 2 September 2025; revised 1 February 2026; accepted 5 March 2026. Date of publication 10 March 2026; date of current version 23 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62471235, in part by the Project funded by China Postdoctoral Science Foundation under Grant 2023M731781, in part by the Open Research Fund of State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University under Grant 24R02, in part by the Key Laboratory of Radar Imaging and Microwave Photonics, Nanjing University of Aeronautics and Astronautics, Ministry of Education under Grant NJ20230005, in part by Jiangsu Geological Bureau Research Project under Grant 2022KY12, Grant 2023KY1, and Grant 2024KJ13, and in part by the Frontier Technologies Research and Development Program of Jiangsu under Grant BF2024070. *(Corresponding authors: Kang Ni; Liang Xiao.)*

Yan Huang is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China, and also with the Geological Exploration Technology Institute of Jiangsu Province, Nanjing 210049, China (e-mail: huangyan_jlu@163.com).

Binghua Lu, Zhizhong Zheng, and Kang Ni are with the School of Computer Science, Nanjing University of Posts and Telecommunications, Nanjing 210023, China (e-mail: lubinghua66@gmail.com; zhengzz_js@126.com; tznikang@163.com).

Liang Xiao is with the School of Computer Science and Engineering and the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information, Ministry of Education, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: xiaoliang@mail.njust.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3672610

I. Introduction

With the rapid development of Earth observation-related technologies, the acquisition of multisource remote sensing images has significantly increased. Different types of remote sensing data often exhibit complementary characteristics, making multisource remote sensing image interpretation [1], [2], [3] widely utilized in areas such as military target detection, mineral resource exploration, and agricultural monitoring. At present, the integration of hyperspectral imagery (HSI) and light detection and ranging (LiDAR) for land cover classification has emerged as a key focus of research globally [4], [5]. The fusion of these two datasets combines spectral features, spatial structures, and elevation information [6], [7], [8], offering a multidimensional and comprehensive representation of land covers. This enhances the distinctiveness of object categories, thereby improving classification accuracy.

A. Prior Works on Hyperspectral and LiDAR Classification

Currently, most research on HSI and LiDAR collaborative classification algorithms focuses on the following key issues.

1) The scene of land cover is relatively complex, and hyperspectral sensors can acquire rich spectral and spatial information. It is necessary to effectively capture spectral–spatial feature representations to enhance classification performance.
2) For the same land covers, the statistics provided by HSI and LiDAR data is interrelated and complementary. This complementary and correlated attribute of multimodal data needs to be fully utilized to improve the performance of land cover classification.

Most related improved algorithms based on convolutional neural networks (CNNs) and Transformer utilize attention mechanisms [9], feature interaction, and spectral–spatial feature fusion [10], [11], [12], [13], and so on. Gao et al. [14] developed an adaptive spatial–spectral enhancement network, including texture, spatial structure, and spectral feature learning and fusion, for enhancing the distinguishability of spectral–spatial characteristics of HSI and LiDAR data. Feng et al. [15] addressed the issues of single-scale feature unreliability and the high-dimensional complexity of multiscale features via similarity calculation, and they utilized spatial attention and modal attention to acquire the deep

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

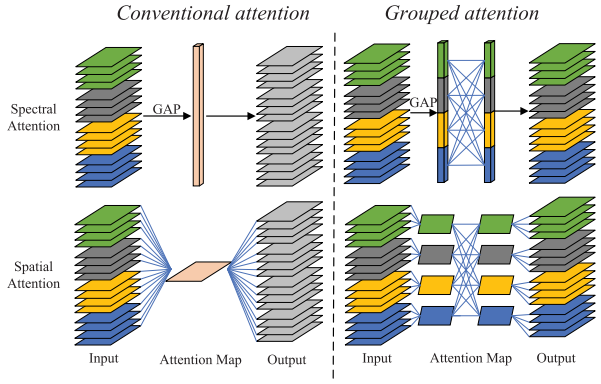


Fig. 1. Conventional attention and grouped attention. GAP denotes the global average pooling layer.

joint features. Zhao et al. [16] concentrated on feature learning with a dual-branch spectral–spatial Transformer and a dynamic adaptive feature fusion mechanism, applying these methods to the collaborative classification of HSI and LiDAR, and achieved promising classification performance. All of the above algorithms are based on the CNNs-Transformer framework and follow a process of feature fusion and classification from dual-branch spectral–spatial feature learning. These algorithms can effectively enhance the classification performance of HSI and LiDAR land covers [17] by learning from local–global features, spectral–spatial features, and feature interactions, but they still have some limitations.

1) *Local Correlations and Prior Information:* For spectral–spatial feature learning, the majority of algorithms utilize global average pooling (GAP)-based global attention mechanisms, neglecting the learning of local spectral and spatial correlations in HSI data. Furthermore, the low-rank prior information of HSI data has not yet been considered in their classification tasks.
2) *Local Attributes of Feature Fusion:* Global feature fusion could capture the overall distribution patterns of different land-covers. However, for land-cover types with similar global features but different local details, such as trees and shrubs, local attributes can provide additional discriminative information. Therefore, integrating local attributes into the learning process of global feature fusion thus facilitates a more comprehensive final representation of land-covers.

B. Motivation

Currently, the separability of group convolution enables each group to focus on different local features, reducing interference between groups. This characteristic helps the model capture finer-grained local features, thereby enhancing the diversity of feature representation. In the characterization of HSI and LiDAR land cover features, the grouping mechanism can effectively account for the local correlations of HSI and LiDAR data, capturing features of related spectral or spatial regions and focusing on local similarities. However, grouped convolution has yet to address the redundancy inherent in HSI spectral information.

The attention mechanism plays a pivotal role in fusing HSI and LiDAR data. It capitalizes on the inherent complementarity between these modalities through dynamic, adaptive modeling, thereby significantly improving collaborative classification accuracy. By utilizing attention weights, redundant spectral information can be suppressed, and important spectral features or regions can be highlighted, while pinpointing pixel areas associated with target ground objects in the spatial dimension. Inspired by grouped convolution and attention mechanisms, employing a grouped attention mechanism enables the model to emphasize local features while effectively capturing the correlations between spectral and spatial relationships, as shown in Fig. 1.

Inspired by the grouped feature learning mechanism, we design a grouped mechanism embedded in a CNNs-Transformer-based HSI and LiDAR classification network, named grouped attention-based spectral–spatial feature learning (GAS^2-FL) network. Different from other algorithms, the proposed grouped attention mechanism models the correlation between spectral and spatial domains while fully considering the low-rank prior of HSI data. To further exploit the relationships between heterogeneous features, we incorporate a nonlocal enhancement model. This module is designed to consolidate features from various local regions, thereby empowering the feature fusion network to capture more discriminative representations equipping with the grouped mechanism. In summary, this article makes the following key contributions.

1) We design a grouped enhanced spectral–spatial feature learning module, which includes a HSI-guided grouped spectral interaction feature learning (GSpectral-IFL) module and a HSI-guided grouped spatial feature learning module (GSpatialFL). The GSpectral-IFL leverages spectral interaction attention and low-rank prior to model the spectral information interaction between different features and enhance the spectral features.
2) We design a grouped attention mechanism for each group focusing on different features or structures, enabling the model to capture more powerful feature representations in GSpatialFL. Specifically, GSpatialFL still relies on the low-rank prior of HSI in the HSI feature learning branch.
3) We integrate the grouped token mechanism and nonlocal token learning mechanism into the global feature fusion learning module, proposing the grouped pointwise transformer (GPTransformer) module. This module captures the global fusion feature statistics of land covers while considering local feature properties and interactions.

The proposed GAS^2-FL model was evaluated on four HSI and LiDAR fusion datasets, where it demonstrated outstanding classification accuracy and robust performance even under limited training samples.

II. Related Work

A. Spectral–Spatial Feature Learning

HSI and LiDAR data exhibit highly complementary characteristics and combining spatial and spectral feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

learning [18], [19] can fully exploit the advantages of these data, thereby significantly enhancing classification performance [6], [20]. Particularly in complex scenes, spectral–spatial joint features can distinguish land covers that are spectrally similar but differ significantly in texture or shape. In recent years, a large number of deep learning networks [21], [22] based on spectral–spatial feature interaction learning mechanism have emerged, such as, deep hierarchical vision transformer [16], hierarchical CNNs and transformer [23], convolution interaction transformer [24], mixed CNN and lightweight transformer [25], spectral–spatial-elevation fusion transformer [26], dual-branch feature fusion network [27], spectral–spatial graph network [28], spatial–spectral salient reinforcement network [29], multiscale spatial–spectral enhancement network [14], spatial–spectral cross-modal attention network [30], and others.

At present, most algorithms adopt a dual-branch spectral–spatial feature learning framework, with many adopting CNNs and transformer for local–global feature learning. As modeling global features, embedding the contextual information of local or similar land covers into the Transformer structure could more comprehensively characterize the properties of the land covers. Subsequently, this article employs a grouped mechanism to cluster tokens in the Transformer, thereby representing local or similar objects. Subsequently, it utilizes a nonlocal token learning module and a pointwise transformer module to model more comprehensive land cover attributes.

B. Attention Mechanism

Attention mechanisms can dynamically generate feature weights to enhance feature representation. Currently, the popular attention mechanisms include spatial attention, channel attention, self-attention mechanisms, hybrid attention, and others. Commonly used attention mechanism algorithms include SENet [31], CBAM [32], GCNet [33], ECANet [34], PANet [35], and others. Reasonable utilization of attention mechanisms can effectively enhance HSI and LiDAR feature representations, thereby boosting the performance of land-cover classification, such as, dual attention [36], heterogeneous attention feature fusion [37], attention-aware CNNs [21], attention calibration network [38], multiattentive hierarchical dense fusion [39], multilevel attention dynamic-scale network [40]. In particular, Mei et al. [41] designed a grouped pixel embedding module into hierarchical vision transformer block to model local relationships between different spectral, achieving effective classification results. This algorithm applies the grouping mechanism to HSI land-cover classification. However, the low-rank characteristics of HSI data, spectral correlation modeling, and the relevance of local feature correlations are crucial for characterizing HSI and LiDAR features.

Therefore, we utilize a grouped attention mechanism to model the correlations between spectra of different land cover properties, embedding low-rank priors to effectively capture the spectral characteristics of HSI. Additionally, grouped attention could focus on different spatial features with similar land cover properties, enabling better spatial feature learning for HSI and LiDAR data.

In summary, our contribution does not lie in grouping per se, but in a fusion-oriented local–global interaction mechanism tailored to HSI–LiDAR. We introduce hyperspectral-aware spectral and spatial grouping to capture local structures, represent each group with compact group tokens, and perform an explicit intergroup interaction stage to enable structured message passing across groups. In parallel, a low-rank memory unit is coupled with grouped spectral modeling to strengthen global spectral and channel statistics under limited labeled samples, and a nonlocal fusion module is incorporated to model long-range dependencies and enhance cross-modality coupling. In contrast to grouped pixel embeddings and region-token ViTs that primarily employ grouping for locality or efficiency and typically restrict interactions to within-group attention, our framework establishes a principled information flow that integrates local modeling with explicit global reasoning for robust multimodal fusion.

III. Methodology

Designed for pixel-level land use and land cover (LULC) classification, our GAS^2-FL model tackles the challenge by synergistically utilizing HSI and LiDAR data. The overall architecture of the proposed framework is depicted in Fig. 2.

A. Overview

Our GAS^2-FL is composed of four parts: GSpectral-IFL, GSpatialFL, LiDAR-guided spatial feature learning (LiDAR-SFL) module,, and GPTransformer. By applying a 3-D convolution operation to HSI data and dividing the spectra into groups, GSpectral-IFL and GSpatialFL extract spatial and spectral features from HSI data through spatial and spectral attention mechanisms. The LiDAR-SFL employs spatial attention by considering grouped information to capture spatial features and uses a channel convolution operation to aggregate diverse spatial information. GPTransformer, building upon the nonlocal joint fusion module (NLFusion), learns a graph awareness through spatially grouped data, thereby balancing global and local receptive fields. Finally, the token fusion module [42] facilitates the fusion of two-modality data and leads to enhanced performance.

In our GAS^2-FL, the input HSI data is $\mathbf{X_{HSI}} \in \mathbb{R}^{p\times p\times s_h}$ and input LiDAR data is $\mathbf{X_{LiDAR}} \in \mathbb{R}^{p\times p}$, where p is the patch size parameter and s_h is the spectrum which is determined by the dataset. So, we first employ a convolution operation for spectrum control and elementary feature extraction. To effectively extract spectral–spatial features, the HSI convolution result is partitioned into N_g groups. Subsequently, a 3-D convolution operation is employed on grouped data $\mathbf{X_g} \in \mathbb{R}^{N_g\times C/(N_g)\times p\times p}$ to generate additional heads for capturing more effective statistics, where C is the channel dimension after the initial convolution. The 3-D convolution result is $\mathbf{X_h} \in \mathbb{R}^{h\times N_g\times C/(N_g)\times p\times p}$, where h denotes the number of heads the data would be partitioned into. Each head functions as a subordinate unit for subsequent GSpectral-IFL and GSpatialFL. After GSpectral-IFL and GSpatialFL, with sufficient spectral–spatial information, the comprehensive information is aggregated by convolutions. Due to the absence of spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

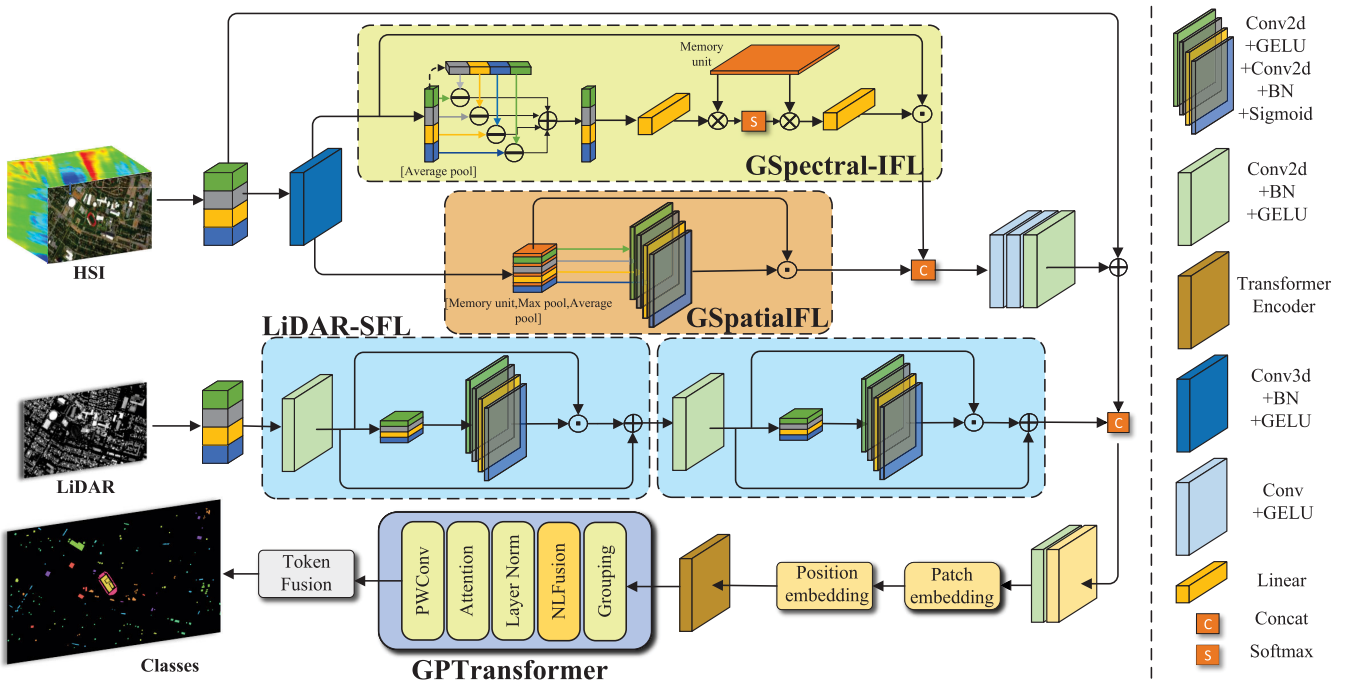


Fig. 2. The overview of our GAS^2-FL. It highlights four key components, i.e., GSpectral-IFL, GSpatialFL, LiDAR-SFL, and GPTransformer. "Conv2d" and "Conv3d" denote a 2-D and a 3-D convolution operation. "BN" is batch normalization. "GELU" and "Sigmoid" represent two different activation functions. "Linear" is a linear mapping operation.

information in LiDAR data, LiDAR-SFL is employed on LiDAR data to achieve spatial enhancement. Subsequently, we concatenate two datasets and apply two convolutional layers: one to reduce the patch size for the next spatial partitioning and the other for data integration. Afterward, patch embedding and position embedding are employed for the Transformer encoder. Then our GPTransformer is employed to capture the global features while considering local feature properties. The NLFusion module bridges local and global feature learning through graph convolution.

B. HSI-Guided GSpectral-IFL Module

The GSpectral-IFL is designed to enhance feature accuracy and robustness via grouped spectral interaction. To obtain discriminative spectral descriptors, we derive a compact vector representation and use it to recalibrate spectral responses, following [31]. For clarity, we describe GSpectral-IFL (and GSpatialFL) within the head-wise feature tensor $\mathbf{Z_h} \in \mathbb{R}^{(hN_g)\times C/(N_g)\times p\times p}$, where p is the patch size, N_g is the number of spectral groups, and h is the number of heads produced by the 3-D convolution. We reshape $\mathbf{Z_h}$ into $\mathbf{Z} \in \mathbb{R}^{N_g\times h\times C/(N_g)\times p\times p}$ to explicitly index groups and heads, and denote the feature map of the ith group as $\mathbf{G}_i \in \mathbb{R}^{h\times C/(N_g)\times p\times p}$.

For each head slice, we apply GAP $GAP(\cdot)$ over the spatial dimensions. Specifically, for any feature map $\mathbf{U} \in \mathbb{R}^{C/(N_g)\times p\times p}$, we have $GAP(\mathbf{U}) \in \mathbb{R}^{C/(N_g)}$. Following [43], the spectral descriptor of the ith group is calibrated by intergroup differences

$$\mathbf{f_{CAi}} = \frac{1}{N_g - 1}\phi\left(\sum_{\substack{n=1\\ n\neq i}}^{N_g} GAP\left(\mathbf{G}_i - \mathbf{G}_n\right)\right) \tag{1}$$

where $\mathbf{f_{CAi}} \in \mathbb{R}^{C/(N_g)}$ is the spectral descriptor of the ith group, and $\phi(\cdot)$ denotes a channel-wise gating mapping implemented by a 1×1 convolution (or an equivalent linear layer) followed by a Sigmoid activation. Equation (1) explicitly emphasizes inter-group discrepancies to strengthen discriminative spectral cues.

Afterward, the group descriptors are concatenated along the spectral/channel dimension to obtain the head-level descriptor: $\mathbf{f_{CA}} = [\mathbf{f_{CA1}}; \mathbf{f_{CA2}}; \cdots; \mathbf{f_{CA_{Ng}}}]$, thus yielding $\mathbf{f_{CA}} \in \mathbb{R}^C$.

Most existing approaches for joint land-cover classification using HSI and LiDAR rely on small isolated patches, which limits global contextual awareness. To alleviate this issue, we introduce a memory unit to aggregate global features within both spectral and spatial attention modules, following [44]. Concretely, we adopt a rank-bounded memory bank $\mathbf{M} \in \mathbb{R}^{K\times B}$ as learnable parameters, where K is the latent subspace dimension and B is the number of memory atoms.

The *low-rank* property is enforced by explicit rank parameterization through the subspace dimension K (with $K \ll C/(N_g)$), rather than by an additional SVD/nuclear-norm regularizer. Specifically, for each of the (hN_g) slices in $\mathbf{Z_h}$, we compute a channel descriptor and project it to a K-dimensional subspace using $\mathbf{W_s} \in \mathbb{R}^{C/(N_g)\times K}$. The memory bank stores B learnable atoms within this K-dimensional subspace. In practice, we set $K = 1/4C/(N_g)$ and $B = 128$ unless otherwise specified.

Let $\mathbf{F_{CA}} \in \mathbb{R}^{(hN_g)\times C/(N_g)}$ denote the slice-wise channel descriptors used to query the memory unit (one descriptor per slice). We compute the memory coefficients $\mathbf{I} \in \mathbb{R}^{(hN_g)\times B}$ by

$$\mathbf{I} = \text{Softmax}\left(\mathbf{F_{CA}}\mathbf{W_s}\mathbf{M}\right) \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

where Softmax(·) is applied along the memory-slot dimension B (i.e., over the last dimension) such that each row satisfies $\sum_{b=1}^{B} I_{t,b} = 1$. Equation (2) performs a content-based addressing over B memory atoms.

Given $\mathbf{I}$, we retrieve global statistics in the same subspace and lift them back to the channel space using $\mathbf{W_e} \in \mathbb{R}^{K\times C/(N_g)}$

$$\mathbf{f_{att}} = \text{Sigmoid}\left(\mathbf{I}\mathbf{M}^{\top}\mathbf{W_e}\right) \tag{3}$$

where $\mathbf{f_{att}} \in \mathbb{R}^{(hN_g)\times C/(N_g)}$ provides slice-wise channel gates. Finally, $\mathbf{f_{att}}$ is broadcast over the spatial dimensions and applied to $\mathbf{Z_h}$ by element-wise multiplication

$$\mathbf{Z_{CA}} = \mathbf{Z_h} \odot \mathbf{f_{att}} \tag{4}$$

where $\odot$ denotes element-wise product with broadcasting over the spatial dimensions. Equations (3) and (4) integrate global memory cues into grouped spectral features.

C. HSI-Guided GSpatialFL

To extract spatial information of the input $\mathbf{Z}$ within each group, both average pooling and max pooling are utilized, with the objective of generating a weight map that is subsequently applied through multiplication, as referenced by [32]. Similar to the operation in spectral attention, a memory unit is integrated within each group to aggregate global information.

To acquire an enhanced attention map, two convolutional layers followed by a sigmoid activation function are applied to produce it

$$\mathbf{f_t} = GELU\left(Conv_{3\times3}\left(\left[\mathbf{M_1};\mathbf{G_{avg1}};\mathbf{G_{max1}};\ldots;\mathbf{G_{maxi}}\right]\right)\right) \tag{5}$$

$$\mathbf{f_{att}} = Sigmoid\left(Conv_{3\times3}\left(\mathbf{f_t}\right)\right) \tag{6}$$

where GELU represents the activation function, Gaussian error linear unit (GELU), and within each group, $\mathbf{M}_i$ denotes the memory unit, $\mathbf{G_{avg}}_i$ represents the average pool product of each group, and $\mathbf{G_{max}}_i$ corresponds to the max pooled output of the group. *Conv* is the convolution operation, with the subscript 3×3 indicating the kernel size. The memory unit is shared across different heads to facilitate more effective optimization, serving as a representation of the global information.

The output of the spatial module is given by the following equation:

$$\mathbf{Z_{SA}} = \left[\mathbf{G_1} \odot \mathbf{f_{att1}};\mathbf{G_2} \odot \mathbf{f_{att2}};\ldots;\mathbf{G_{N_g}} \odot \mathbf{f_{attN_g}}\right] \tag{7}$$

where f_{atti} represents the ith channel of f_{att}.

Finally, each head aggregates independent spectral and spatial information. The spectral and spatial features are integrated by concatenation, followed by hierarchical convolutional operations for further aggregation. Several convolutional blocks effectively fuse the spectral–spatial information embedded within the data. Then $\mathbf{Z}_i$ is the ith head, a residual connection is incorporated to enhance performance, as shown in the following equation:

$$\mathbf{Z_{out}} = Conv\left[\mathbf{Z_1};\mathbf{Z_2};\mathbf{Z_3};\ldots;\mathbf{Z_{N_g}}\right] + \mathbf{Z_{in}}. \tag{8}$$

D. LiDAR-SFL Module

As mentioned in [42], the separable convolution layer demonstrates superior performance in feature extraction from LiDAR data compared to the standard convolution layer. Accordingly, we design the LiDAR-SFL in a manner analogous to the separable convolution layer. Similar to GSpatialFL employed in HSI (without the low-rank unit), we capture spatial features through grouped processing and partition the input data along the channel dimension to obtain $\mathbf{X}_{l,in} \in \mathbb{R}^{N_g\times s/(N_g)\times p\times p}$. The attention map is generated by two convolutional layers with a Sigmoid activation

$$\mathbf{f_t} = GELU\left(Conv_{3\times3}\left(\left[\mathbf{G_{avg1}};\mathbf{G_{max1}};\ldots;\mathbf{G_{maxi}}\right]\right)\right) \tag{9}$$

$$\mathbf{f_{att}} = Sigmoid\left(Conv_{3\times3}\left(\mathbf{f_t}\right)\right) \tag{10}$$

where $\mathbf{G_{avg}}$ and $\mathbf{G_{max}}$ are the average pooling and max pooling results, respectively. Cross-group feature fusion is accomplished through intergroup connectivity integrated with convolutional operations. The resulting attention map is broadcast and applied to the input feature to obtain the spatially enhanced representation $\mathbf{X}_{SA}$.

Then we perform further feature extraction by a pointwise convolution

$$\mathbf{X}_{SA} = GELU\left(BN\left(Conv_{1\times1}\left(\mathbf{X}_{SA}\right)\right)\right) \tag{11}$$

as defined in (11). Lastly, a residual connection is employed to acquire the output

$$\mathbf{X}_{out} = \mathbf{X}_{SA} + \mathbf{X}_{l,in}. \tag{12}$$

The two modalities are concatenated along the spectral dimension following the aforementioned feature learning procedure. They are then fed into a 3×3 convolutional layer without padding, which reduces the patch size for the subsequent GPTransformer. After this reduction, a 1×1 convolution is applied to adjust the spectral dimension and facilitate cross-modality information integration. This is followed by positional embedding, patch embedding, and the transformer encoder.

Before GPTransformer, we employ a convolutional downsampling projection with *zero padding* to obtain a compact spatial grid for subsequent grouped processing. Although zero padding could potentially introduce border artifacts, we inspect the corresponding error maps in the qualitative analysis and do not observe systematic error accumulation near image boundaries. This suggests that the proposed downsampling does not introduce noticeable boundary bias.

E. NLFusion Module

We propose an enhanced self-attention mechanism, termed the nonlocal joint feature fusion (NLFusion) module, for joint HSI and LiDAR feature extraction (Fig. 3). NLFusion introduces a lightweight gating-and-pooling strategy to construct a compact nonlocal graph, followed by graph reasoning and a residual fusion back to the token domain.

Let the input token sequence be $\mathbf{X}_{in} \in \mathbb{R}^{N\times C}$, where N is the token number ($N = p^2$ for a $p \times p$ patch) and C is the feature dimension. We reshape features into H heads with per-head

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

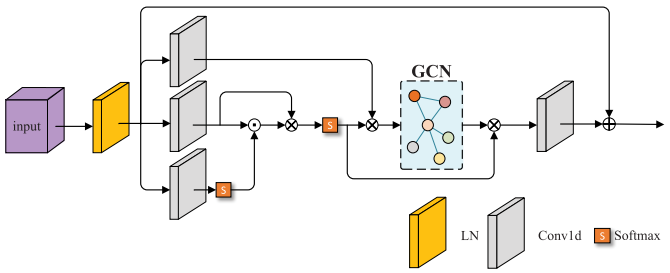


Fig. 3. Architecture of the NLFusion module. "LN" is layer normalization. "GCN" represents a graph convolution. "Conv1d" is a 1-D convolution operation.

dimension d (typically $Hd \approx C/2$ after channel reduction). For simplicity, we write the multihead tensors as $\mathbf{X}_Q, \mathbf{X}_K \in \mathbb{R}^{H\times d\times N}$

$$\begin{cases} \mathbf{X}_K = \mathbf{W}_K \mathbf{X}_{in}, & \mathbf{X}_K \in \mathbb{R}^{H\times d\times N} \\ \mathbf{X}_Q = \mathbf{W}_Q \mathbf{X}_{in}, & \mathbf{X}_Q \in \mathbb{R}^{H\times d\times N} \\ \mathbf{a} = \left(\text{Softmax}_d\left(\mathbf{X}_Q\right)\right)_{[r]}, & \mathbf{a} \in \mathbb{R}^{H\times 1\times N} \\ \mathbf{X}_V = P\left(\mathbf{X}_K \odot \mathbf{a}\right), & \mathbf{X}_V \in \mathbb{R}^{H\times d\times m} \\ \mathbf{X}_{att} = \text{Softmax}_m\left(\mathbf{X}_V^{\top}\mathbf{X}_K\right), & \mathbf{X}_{att} \in \mathbb{R}^{H\times m\times N}. \end{cases} \tag{13}$$

In (13), $\odot$ denotes element-wise multiplication with broadcasting along the channel dimension. The pooling operator $P(\cdot)$ is defined as adaptive average pooling along the token axis

$$P: \ \mathbb{R}^{H\times d\times N} \to \mathbb{R}^{H\times d\times m}. \tag{14}$$

We apply two Softmax operations with different roles in (13): $\text{Softmax}_d(\cdot)$ normalizes along the channel/state dimension d, and $(\cdot)_{[r]}$ selects one normalized channel (fixed index r) to form the gating vector $\mathbf{a}$; $\text{Softmax}_m(\cdot)$ normalizes along the pooled-node dimension m, producing the final attention projection $\mathbf{X}_{att}$.

Using the attention projection $\mathbf{X}_{att}$ defined in (13), we project token features into the pooled graph domain [45] as in (15), where $\mathbf{X}_{att}^T$ aligns the token axis (N) to the pooled-node axis (m). We then apply a single-layer GCN over the pooled nodes to capture nonlocal relations, as formulated in (16)

$$\mathbf{X}_g = \mathbf{X}_K \mathbf{X}_{att}^T \in \mathbb{R}^{H\times d\times m} \tag{15}$$

where $\mathbf{X}_{att}^T \in \mathbb{R}^{H\times N\times m}$. A single-layer GCN is applied to capture nonlocal relations among pooled nodes

$$\hat{\mathbf{X}}_g = \text{ReLU}\left(\left(\mathbf{I} - \mathbf{A}\right)\mathbf{X}_g \mathbf{W}_g\right) \tag{16}$$

where $\mathbf{A} \in \mathbb{R}^{m\times m}$ is the adjacency matrix, $\mathbf{I}$ is the identity matrix, and $\mathbf{W}_g$ is the learnable projection.

Finally, we map the graph-enhanced representation back to the token domain by (17) and fuse it with the original tokens via a residual connection in (18). The pooling operator $P(\cdot)$ used to obtain $\mathbf{X}_V$ is defined in (14), ensuring that the nonlocal reasoning is performed on a compact set of m pooled nodes rather than all N tokens

$$\mathbf{X}_{rec} = \hat{\mathbf{X}}_g \mathbf{X}_{att} \in \mathbb{R}^{H\times d\times N} \tag{17}$$

$$\mathbf{X}_{out} = \mathbf{X}_{in} + \mathbf{W}_O \ \text{Concat}\left(\mathbf{X}_{rec}\right) \in \mathbb{R}^{N\times C} \tag{18}$$

where Concat$(\cdot)$ concatenates the H heads along the channel dimension and $\mathbf{W}_O$ is a 1×1 projection.

F. GPTransformer Module

To boost the locality of the standard Transformer model, we adopt a groupwise multihead attention structure, as depicted in Fig. 4. First, the input feature map is partitioned along the spatial dimensions into distinct groups of uniform size. Each group is then transformed into a token sequence and concatenated with a randomly initialized group token, which serves as a representative token encapsulating the primary features of the corresponding group. Subsequently, the proposed NLFusion layer is applied to extract groupwise information

$$\mathbf{X}_t = \text{NLFusion}\left(\text{LN}\left(\text{Concat}\left(\mathbf{X}, \mathbf{gt}\right)\right)\right) \tag{19}$$

where $\mathbf{gt} \in \mathbb{R}^C$ denotes the group token, $\mathbf{X}$ denotes the token sequence of all groups (after spatial partition and flattening), Concat$(\cdot)$ concatenates $\mathbf{gt}$ to each group sequence, and LN$(\cdot)$ represents the layer normalization.

To capture comprehensive patch information, a self-attention layer is further employed across groups using the group tokens. Let $\mathbf{GT} \in \mathbb{R}^{N_s\times C}$ denote the set of group tokens (one token per group) after (19), where N_s is the number of spatial groups. The attention is defined as

$$\begin{cases} \mathbf{Q} = \text{LN}\left(\text{GELU}\left(\mathbf{GT}\right)\right)\mathbf{W}_q \\ \mathbf{K} = \text{LN}\left(\text{GELU}\left(\mathbf{GT}\right)\right)\mathbf{W}_k \\ \mathbf{X}_t' = \text{Softmax}\left(\dfrac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{X} \end{cases} \tag{20}$$

where d_k denotes the feature dimension of $\mathbf{K}$ (per head). Equation (20) enables each group token to aggregate information across groups, thus capturing global interactions between different local regions. As in the Transformer architecture, all operations are performed with n heads, while the group token is shared across heads.

Given the high-level semantic nature of the group-token interactions, a single 1×1 convolution is sufficient to aggregate the multihead global features. The complete GPTransformer output is expressed as

$$\mathbf{X}_T = \left[\mathbf{X}_{t1}'; \mathbf{X}_{t2}'; \cdots; \mathbf{X}_{tn}'\right] \tag{21}$$

$$\mathbf{X}_{out} = \text{Conv}_{1\times 1}\left(\text{LN}\left(\mathbf{X}_T\right)\right) \tag{22}$$

where $[\cdot;\cdot]$ denotes concatenation along the channel dimension.

Finally, $\mathbf{X}_{out}$ is forwarded to the Token Fusion module to derive the final classification. Specifically, the Token Fusion module employs an MLP prehead to project $\mathbf{X}_{out}$ into 1-D token weights, which are then used to compute a weighted sum of token features, followed by an MLP classifier.

IV. Experiments

For assessing the efficacy of GAS^2-FL, the study employs four prevalent HSI and LiDAR datasets.

A. Dataset

1) Houston2013: The Houston2013 dataset includes a hyperspectral image and a LiDAR-based digital surface model (DSM), acquired in June 2012 over the University of Houston and its surrounding urban region in Texas, USA. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

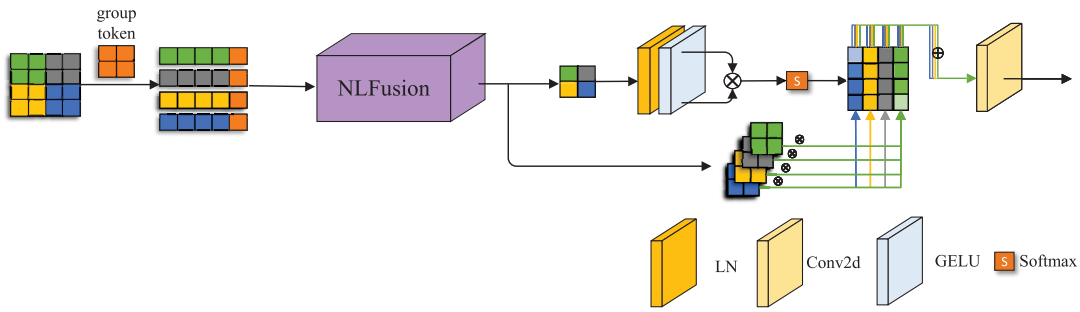


Fig. 4. Architecture of the GPTransformer module.

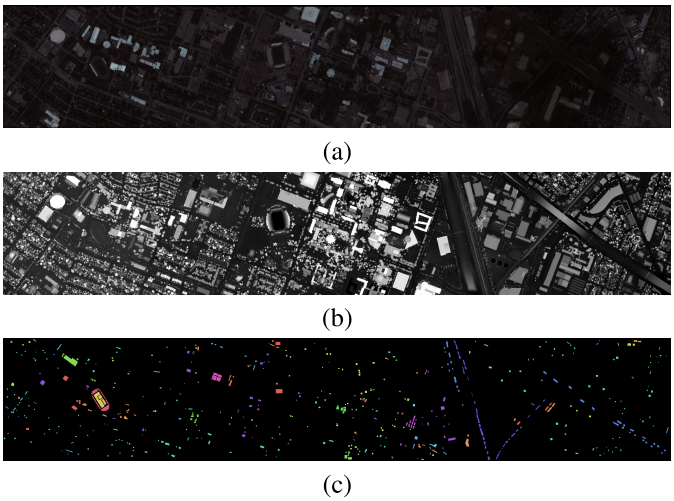


Fig. 5. Illustration of the Houston2013 dataset. (a) Pseudocolor composite image based on bands 10, 20 and 30 for HSI. (b) LiDAR-based DSM. (c) Ground truth.

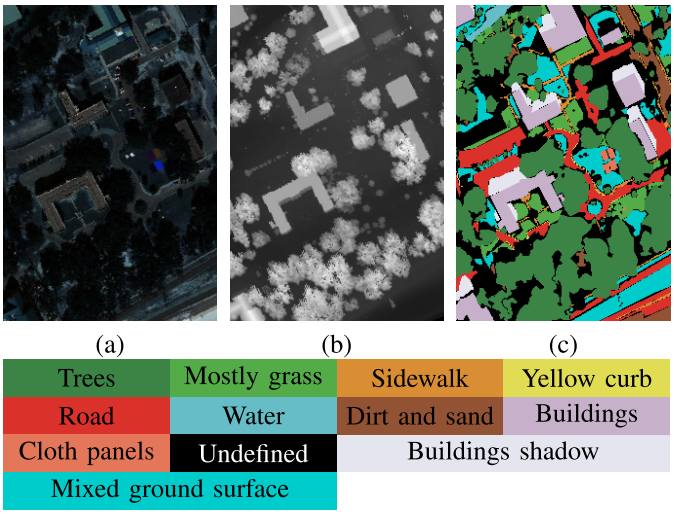


Fig. 6. Illustration of the MUUFL dataset. (a) Pseudocolor composite image based on bands 10, 20 and 30 for HSI. (b) LiDAR-based DSM. (c) Ground truth.

hyperspectral data were captured using the ITRES CASI-1500 sensor operated by the National Center for Airborne Laser Mapping, featuring 144 spectral bands with a resolution of 10 nm across the 380–1050 nm wavelength range. The accompanying LiDAR data offer single-band elevation measurements. Both data sources share a spatial resolution of 2.5 m and cover an area of 349 $\times$ 1905 pixels. Designed for land cover classification tasks, the dataset provides finely annotated ground truth labels for 15 semantic categories, comprising 15 029 validated samples [46]. Fig. 5 displays a pseudo-color representation of the HS image, the LiDAR intensity image, and the reference land cover map.

2) MUUFL: The MUUFL dataset was collected in November 2010 over the Gulf Park campus of the University of Southern Mississippi, situated in Long Beach, Mississippi, USA. It consists of coregistered hyperspectral images and a LiDAR-derived DSM, both offered at a 1-m spatial resolution. The hyperspectral imagery was acquired with the ITRES CASI-1500 imaging sensor. To reduce noise, the first and last eight spectral bands were excluded, yielding a total of 64 usable bands spanning the wavelength interval from 380 to 1050 nm. The LiDAR data, acquired with an ALTM sensor operating at 1.06 μm, comprise two distinct raster layers. A total of 53 687 annotated pixels across 11 land-cover categories are provided for classification tasks [17]. Fig. 6 displays a pseudo-color composite (generated from HSI bands 10, 20, and 30), the LiDAR intensity image, and the reference ground-truth labels.

3) Yancheng: The Yancheng dataset encompasses HSI data and a LiDAR-based DSM, acquired on October 15, 2022, over Yancheng, China, at an altitude of 600 m. The HSI data was captured via a visible light imaging spectrometer within the multiparameter airborne remote sensing system (MARSS), containing 249 spectral bands (400–1000 nm). Principal component analysis (PCA) reduced the spectral dimensionality to 30 components to minimize redundancy and computational load. LiDAR data was obtained using a RIEGL VUX-240 scanner operating in a single spectral band, achieving an average point density of 51.44 pts/m^2. Both datasets share a spatial extent of 400 $\times$ 500 pixels with 1-m resolution. Nine land cover categories are represented in 2 00 000 annotated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

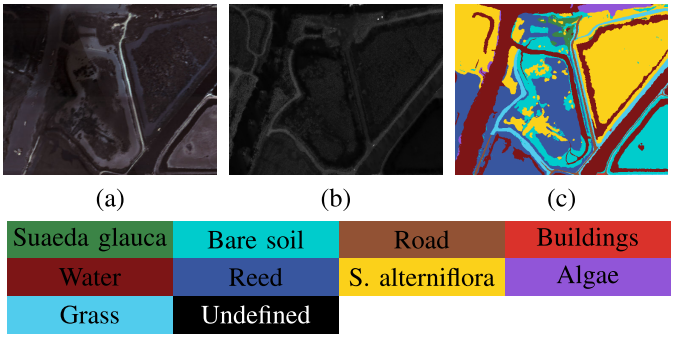


Fig. 7. Illustration of the Yancheng dataset. (a) Pseudocolor composite image based on bands 10, 20, and 30 for HSI. (b) LiDAR-based DSM. (c) Ground truth.

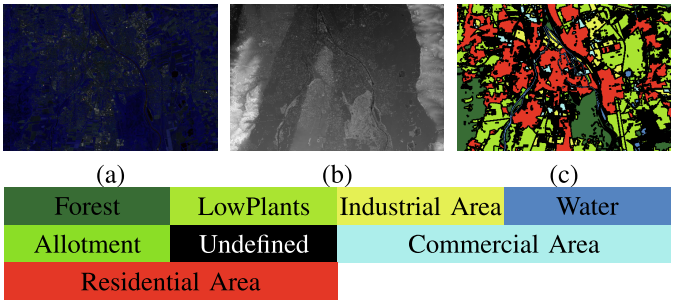


Fig. 8. Illustration of the Augsburg dataset. (a) Pseudocolor composite image based on bands 10, 20 and 30 for HSI. (b) LiDAR-based DSM. (c) Ground truth.

samples [5]. Fig. 7 displays a pseudocolor HSI composite, LiDAR intensity image, and corresponding ground truth.

4) Augsburg: The Augsburg dataset comprises a hyperspectral image captured over an urban region of Augsburg, Germany, with a spatial size of 332 $\times$ 485 pixels and annotations for seven land-cover classes. The hyperspectral data were acquired using the DAS-EOC HySpex sensor, containing 180 spectral bands covering the wavelength range from 0.4 to 2.5 μm. The LiDAR-DSM, acquired using the DLR-3K system, consists of a single-band elevation raster. To enable multimodal analysis, both data sources were resampled to a consistent spatial resolution of 30 m [23]. A pseudocolor composite of the HSI, a grayscale rendering of the DSM, and the reference ground truth are displayed in Fig. 8.

B. Experimental Setup

Our implementation of GAS^2-FL utilizes the PyTorch framework on a workstation equipped with 32 GB RAM and an NVIDIA GeForce RTX 4090 GPU. Network optimization employs the Adam optimizer with a batch size of 80 and is conducted for 500 epochs. The initial learning rate is set to 6×10^{-5} with L_2 regularization ($\lambda = 2 \times 10^{-4}$). We adopt a step-wise learning rate scheduler, which decays the learning rate by a factor of 0.98 every 20 epochs. All learnable parameters are initialized using PyTorch default initialization. For model selection, we choose the checkpoint that achieves the best validation OA. As data preprocessing, band-wise normalization is applied to the spectral channels. Input patches are fixed at 11 $\times$ 11 pixels. Based on the model parameter

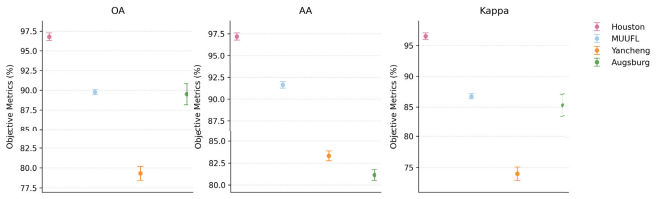


Fig. 9. Statistical validation of GAS^2-FL on four datasets over five independent runs. Error bars denote 95% CIs (95% CI) computed from the Student's t distribution.

configuration, the heads and groups of GSpectral-IFL and GSpatialFL are set to 36 and 4. The depth of LiDAR-SFL is set to 2. Our transformer architecture employs 64-dimensional hidden states, 3 parallel attention heads, and a dropout ratio of 0.5 across all encoder layers. The settings of GPTransformer are as follows: the group size of 3 $\times$ 3 and heads of 7. In each experiment, we construct the training set via stratified per-class sampling: we randomly select 20 labeled samples per class for training on all datasets, except MUUFL where 60 samples per class are used. All remaining labeled samples are used for testing. Unless otherwise stated, we use a fixed random seed of 1 for both network training and testing to ensure reproducibility. The evaluation of classification performance is quantified through three widely adopted metrics: the overall accuracy (OA), average accuracy (AA), and Kappa coefficient.

To assess statistical reliability, we repeat each experiment five times with different random seeds while keeping all hyperparameters fixed. We report the mean and standard deviation (mean $\pm$ std) of OA, AA, and Kappa across runs. Furthermore, we compute the 95% confidence interval (CI) using the Student's t-distribution: $\mu \pm t_{0.975,n-1} \cdot \sigma/\sqrt{n}$ with $n = 5$, where μ and σ denote the sample mean and standard deviation, respectively. Overall, the relatively small standard deviations across runs indicate that our model exhibits stable and reproducible performance under different random seeds. These statistics are used to generate the error bars in Fig. 9.

C. Performance Comparison

The study contrasted the proposed network with leading HSI and LiDAR classification techniques to illustrate its enhanced performance.

To ensure a fair comparison, all methods (GAS^2-FL and baselines) are trained and evaluated under the same train/test split and the same number of training samples per class on each dataset. For each baseline, we rerun its official open-source implementation on our data split, following the original paper's preprocessing procedures and using the same training and evaluation code. Regarding the input patch size, we adopt the patch size recommended in the corresponding baseline paper to reproduce its best-performing setup.

1) MSA-GCN [47] introduced an interactive enhanced network utilizing MSA and GCN for HSI and LiDAR data classification, demonstrating superior performance.
2) MFT [48] network innovatively integrated an external CLS token from multimodal data sources into the transformer encoder, achieving significant performance gains with less effort.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR HOUSTON2013 DATASET (BEST RESULTS ARE IN BOLD)

HOUSTON2013	MSA-GCN	MICF-NET	M2FNet	ExViT	DSHFNet	MHST	MFT	MS2CANet	GAS^2-FL
Healthy Grass	95.78	99.59	86.19	97.33	87.33	97.16	84.57	90.99	**100**
Stressed Grass	87.52	95.87	97.16	98.38	98.70	98.87	89.71	89.47	**99.03**
Synthetic Grass	99.85	97.64	99.7	95.42	**100**	99.11	99.26	98.97	99.85
Tree	97.55	93.38	96.81	93.55	**99.84**	96.32	98.28	94.28	94.20
Soil	**100**	**100**	99.51	99.92	**100**	91.00	96.48	99.83	**100**
Water	96.07	98.03	98.03	96.39	97.04	93.11	92.13	96.72	**99.67**
Residential	90.30	91.27	**95.51**	89.1	77.80	84.94	80.29	76.92	93.11
Commercial	90.28	90.11	**93.3**	88.32	92.73	90.69	76.31	48.12	**93.3**
Road	76.06	87.58	93.02	95.05	85.55	85.8	84.25	93.18	**96.92**
Highway	97.27	97.85	74.73	97.85	98.34	**99.01**	82.77	73.07	93.37
Railway	96.54	**97.61**	95.72	97.53	95.88	91.03	87.49	93.58	96.87
Parking Lot1	79.23	97.61	98.19	92.17	92.75	92.66	90.19	64.30	**98.68**
Parking Lot2	90.65	94.43	93.76	95.32	95.32	95.32	95.54	**95.55**	**95.55**
Tennis Court	**100**	**100**	**100**	99.26	99.02	**100**	95.83	**100**	**100**
Running Track	99.84	**100**	99.69	**100**	99.84	**100**	96.72	**100**	**100**
OA	92.17	95.58	93.95	95.35	93.81	94.36	88.61	85.81	**96.98**
AA	93.13	96.06	94.76	95.73	94.68	94.93	89.99	88.22	**97.37**
Kappa	91.53	95.22	93.47	94.97	93.31	93.91	87.69	84.67	**96.73**

3) MS2CANet [30] employed multiscale pyramid convolutions and EFR modules to extract and enhance features at multiple scales while implementing cross-modal information augmentation.
4) The goal of MHST [49] is to learn informative and efficient features from HSI and LiDAR data by capturing essential land-cover characteristics across varied scales. To this end, the model employs an adaptive mechanism for global feature extraction, which improves efficiency and minimizes redundant computations by activating transformer heads in a selective manner.
5) ExViT [42] incorporated separable convolutional layers, subspace projections, and shared positional embeddings to extract and align both modality-specific and cross-modality features. By integrating hierarchical fusion with classification token fusion, ExViT effectively utilized the spatial ViT backbone to model long-range correlations across modalities.
6) DSHFNet [15] employed distinct attention modules for hierarchical fusion, thereby ensuring comprehensive feature complementarity.
7) M2FNet [25] employs a dual-branch architecture that cleverly blends CNN-driven feature extraction with transformer-based representation learning. It harnesses a multiscale 3-D-2-D hybrid CNN, paired with a dilated convolution component, to efficiently extract and amalgamate key features.
8) MICF-Net [50] leveraged the inherently consistent spatial relationships and high-level semantic information present in multimodal data to guide the extraction of fused multimodal features through hierarchical fusion modules.

We conduct comparisons of different methods' classification performance on the Houston dataset, as shown in Table I, achieving the performance with OA = 96.98%, AA = 97.37%, and Kappa = 96.73%. The proposed network demonstrates perfect classification (100%) in four different categories: Healthy Grass, Soil, Tennis Court, and Running Track, while attaining 99.03% accuracy for Stressed Grass and 99.67% for Water, indicating effective modeling of spectral variations. Spatial discrimination capabilities are evidenced by 96.92% Road accuracy and 93.11% Residential accuracy, outperforming all baselines in complex urban patterns. Though slightly lower accuracy in Tree and Parking Lot2, the 10/15 category-leading performance and 11.17% OA advantage over the weakest baseline confirm holistic context modeling capacity.

On the MUUFL dataset (Table II), our method achieves best performance with OA = 89.86%, AA = 91.25% and Kappa = 86.81%. The observed improvements in urban categories—particularly Building shadow (95.81%) and Road (88.8%)—reflect enhanced spatial–spectral discrimination capabilities. Notably, our framework attains 100% accuracy in Water classification while demonstrating robust performance in spectrally ambiguous classes like Mostly grass (89.14%). Though slight gaps exist in Cloth panels (99.04%) and Yellow curb (91.06%), the comprehensive OA advantage validates the method's urban scene adaptability.

On the Yancheng dataset (Table III), our method achieves OA = 80.44%, AA = 83.44%, and Kappa = 75.37%, demonstrating superior performance in a complex coastal region. The framework demonstrates exceptional performance in similar vegetation categories, achieving accuracies of 94.01% for Suaeda glauca and 83.50% for S. alterniflora, while also exhibiting strong capability in distinguishing substrate types such as Bare soil (79.00%) and Algae (82.15%). Exceptional spectral–spatial information capture capability is evidenced by 100% accuracy on the Buildings class and 84.59% accuracy on the Water class. Although performance gaps are observed in the Road (87.96%) and Reed (74.55%) categories, the 3.24% improvement in OA over the best baseline underscores the framework's comprehensive scene modeling capabilities. Notably, the proposed method attains accuracies exceeding 83% in 5 out of nine categories, substantially outperforming baseline models in challenging scenes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

TABLE II

COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR MUUFL DATASET (BEST RESULTS ARE IN BOLD)

MUUFL	MSA-GCN	MICF-Net	M2FNet	ExViT	DSHFNet	MHST	MFT	MS2CANet	GAS^2-FL
Trees	90.85	90.28	91.19	90.46	91.49	88.74	89.09	86.31	**91.93**
Mostly grass	84.42	80.81	78.76	81.24	81.76	81.66	67.22	76.06	**89.14**
Mixed ground surface	76.08	**83.2**	75.97	79.68	75.74	74.38	77.91	77.44	78.01
Dirt and sand	93.32	**97.33**	90.66	96.09	90.43	90.94	94.39	92.36	95.58
Road	78.63	87.82	83.69	86.54	82.98	87.46	74.03	79.49	**88.8**
Water	99.75	**100**	99.26	99.51	94.09	99.75	99.51	**100**	**100**
Building shadow	90.38	91.39	89.51	94.34	83.34	91.9	83.71	92.5	**95.81**
Building	93.59	92.07	92.48	91.78	93.77	94.08	87.82	91.86	**94.13**
Sidewalk	52.60	80.23	78.11	72.38	78.19	82.87	61.89	78.04	**80.23**
Yellow curb	86.99	**92.68**	74.79	85.37	85.41	94.31	78.86	90.24	91.06
Cloth panels	98.09	**99.52**	**99.52**	99.04	97.13	**99.52**	**99.52**	**99.52**	99.04
OA	86.31	88.66	87.10	87.99	87.14	86.99	83.26	84.56	**89.86**
AA	85.86	90.49	86.72	88.77	83.58	89.60	93.09	87.62	**91.25**
Kappa	82.19	85.30	87.24	84.41	83.16	83.21	78.39	80.18	**86.81**

TABLE III

COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR YANCHENG DATASET (BEST RESULTS ARE IN BOLD)

Yancheng	MSA-GCN	MICF-Net	M2FNet	ExViT	DSHFNet	MHST	MFT	MS2CANet	GAS^2-FL
Suaeda glauca	91.90	92.70	90.16	92.06	95.04	82.06	84.68	90.67	**94.01**
Bare soil	68.62	67.08	71.96	69.30	66.94	78.96	62.77	68.10	**79.00**
Road	87.31	92.29	79.94	88.09	**92.39**	89.65	68.81	91.07	87.96
Buildings	98.85	96.55	97.70	**100**	95.40	96.55	**100**	**100**	**100**
Water	**85.84**	80.64	77.39	74.27	85.42	82.16	78.68	82.82	84.59
Reed	38.09	73.45	68.82	58.00	70.94	69.07	**82.95**	67.36	74.55
S. alterniflora	69.15	81.09	72.31	**83.93**	82.79	78.54	75.37	77.80	83.50
Algae	79.16	81.78	78.43	**82.66**	78.61	76.81	78.81	78.24	82.15
Grass	69.08	62.30	66.00	62.33	44.79	66.65	38.28	67.33	**65.23**
OA	67.35	76.60	75.85	72.89	76.81	77.20	74.01	75.20	**80.44**
AA	76.45	80.87	79.19	78.96	79.15	81.16	74.15	80.38	**83.44**
Kappa	59.06	70.55	69.56	65.79	70.70	71.39	66.84	68.92	**75.37**

TABLE IV

COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR AUGSBURG DATASET (BEST RESULTS ARE IN BOLD)

Augsburg	MSA-GCN	MICF-Net	M2FNet	ExViT	DSHFNet	MHST	MFT	MS2CANet	GAS^2-FL
Forest	77.41	96.12	96.23	93.50	**98.78**	93.88	95.40	97.37	96.72
Residential area	47.00	85.82	89.35	86.36	89.62	72.01	47.67	72.56	**95.32**
Industrial area	51.11	**72.02**	57.56	57.14	18.04	21.22	28.69	57.92	58.39
Low plants	50.11	88.77	86.34	84.02	86.50	79.87	65.97	65.52	**91.65**
Allotment	62.34	**96.04**	58.92	63.98	92.79	92.61	90.63	70.99	88.11
Commercial	20.43	49.48	54.34	52.12	66.09	**79.14**	73.48	55.88	64.37
Water	50.66	65.3	**73.18**	72.38	61.32	69.01	62.31	67.28	72.25
OA	53.15	86.85	86.68	84.20	84.56	76.23	62.39	73.24	**91.35**
AA	51.30	79.08	73.70	72.50	72.92	72.54	66.31	69.65	**80.97**
Kappa	41.27	81.78	81.34	77.92	78.35	69.37	52.28	64.53	**87.75**

On the Augsburg dataset (Table IV), our method achieves state-of-the-art performance with OA = 91.35%, AA = 80.97%, and Kappa = 87.75%. The framework demonstrates balanced competence across feature types: Residential area classification attains 95.32% accuracy, while low plants reaches 91.65%. Notably, it achieves near-top performance in water (72.25%) and forest (96.72%), while maintaining a high level of competitiveness in the interpretation of natural features. Challenges arise in the Industrial area (58.39%) and commercial area (64.37%), which constitute the most structurally complex urban categories. The observed 4.67% gain in OA against the top baseline model demonstrates the efficacy of multimodal feature integration. This is further evidenced by 4 out of the seven categories either surpassing or nearly matching state-of-the-art performance, notwithstanding certain localized challenges.

The consistent performance advantages of our method across Houston, MUUFL, Yancheng, and Augsburg datasets stem from the synergistic effects of three modules: The GSpectral-IFL module, through its spectral interaction attention and low-rank constraints, explicitly amplifies discriminative spectral patterns, enabling precise differentiation of challenging categories with subtle spectral differences—such as achieving best accuracy in both Healthy Grass and Stressed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

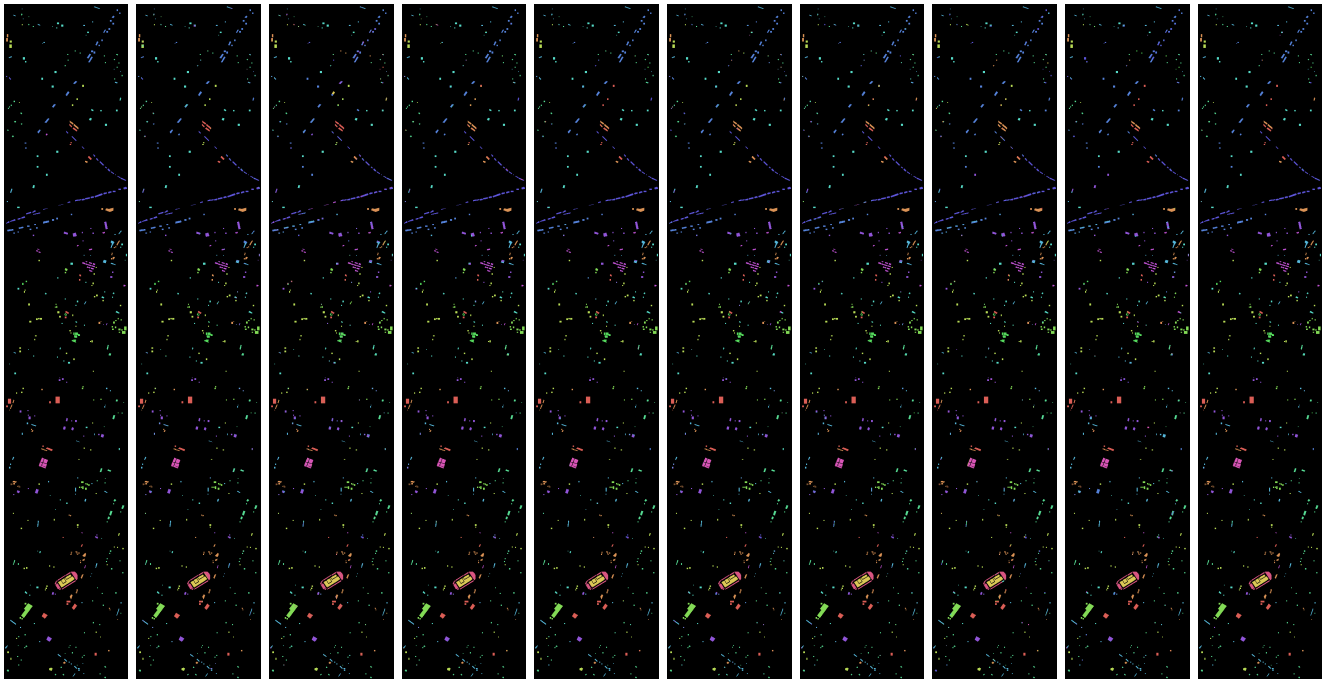


Fig. 10. Visual comparisons on Houston2013 dataset. (a) Ground truth. (b) MSA-GCN(92.17%). (c) MICF-Net(95.58%). (d) M2FNet(93.95%). (e) ExViT(95.35%). (f) DSHFNet(93.81%). (g) MHST(94.36%). (h) MFT(88.61%). (i) MS2CANet(85.81%). (j) GAS^2-FL(96.98%). All maps share the same class-to-color mapping as the Houston2013 ground-truth map in Fig. 5.

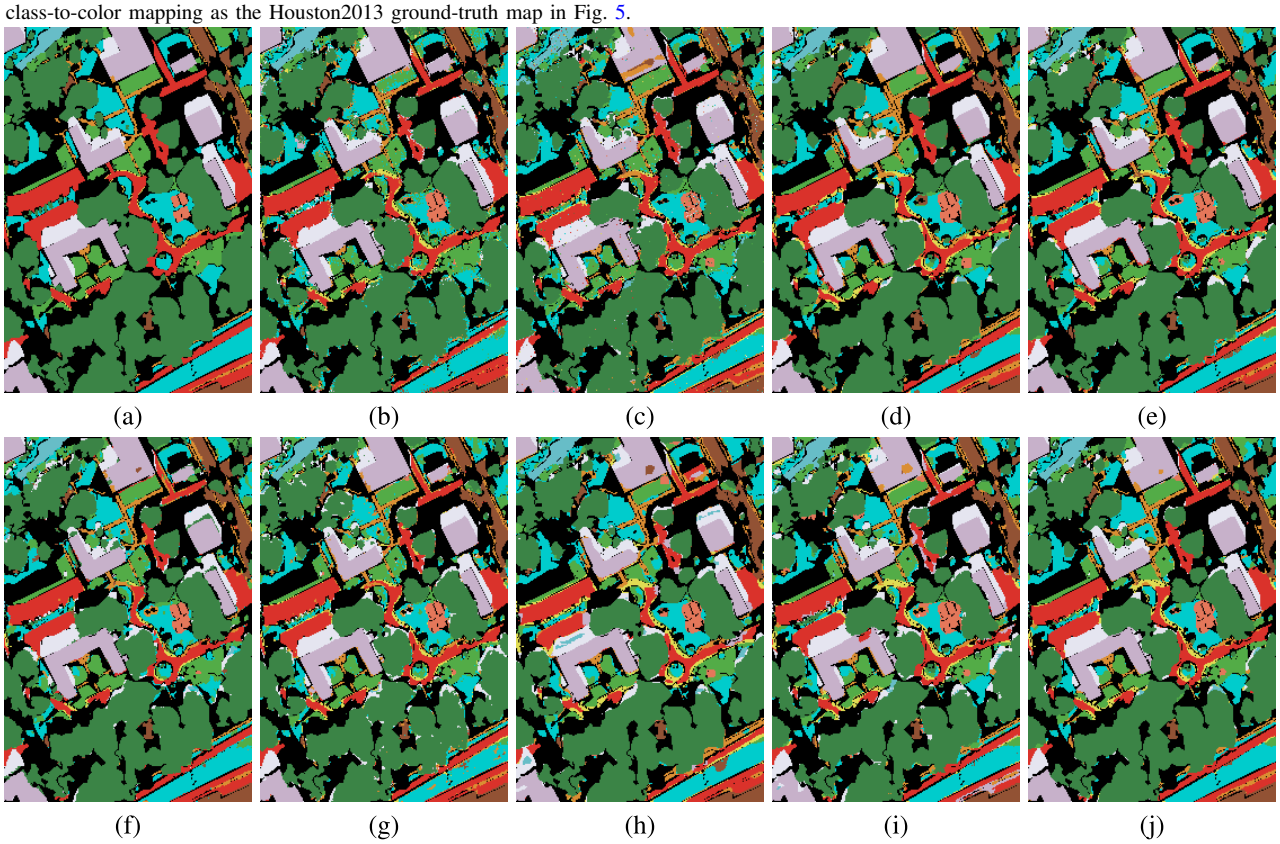


Fig. 11. Visual comparisons on MUUFL dataset. (a) Ground truth. (b) MSA-GCN(86.31%). (c) MICF-Net(88.66%). (d) M2FNet(87.1%). (e) ExViT(87.99%). (f) DSHFNet(87.14%). (g) MHST(86.99%). (h) MFT(83.26%). (i) MS2CANet(84.56%). (j) GAS^2-FL(89.86%). All maps share the same class-to-color mapping as the MUUFL ground-truth map in Fig. 6.

Grass (Houston). Moreover, the GSpatialFL module utilizes adaptive spatial attention to isolate structural patterns per category, resolving intraclass heterogeneity in complex scenes. Finally, the GPTransformer integrates local category-specific features (via grouped tokens) with global contextual distributions (via nonlocal tokens), ensuring balanced performance in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

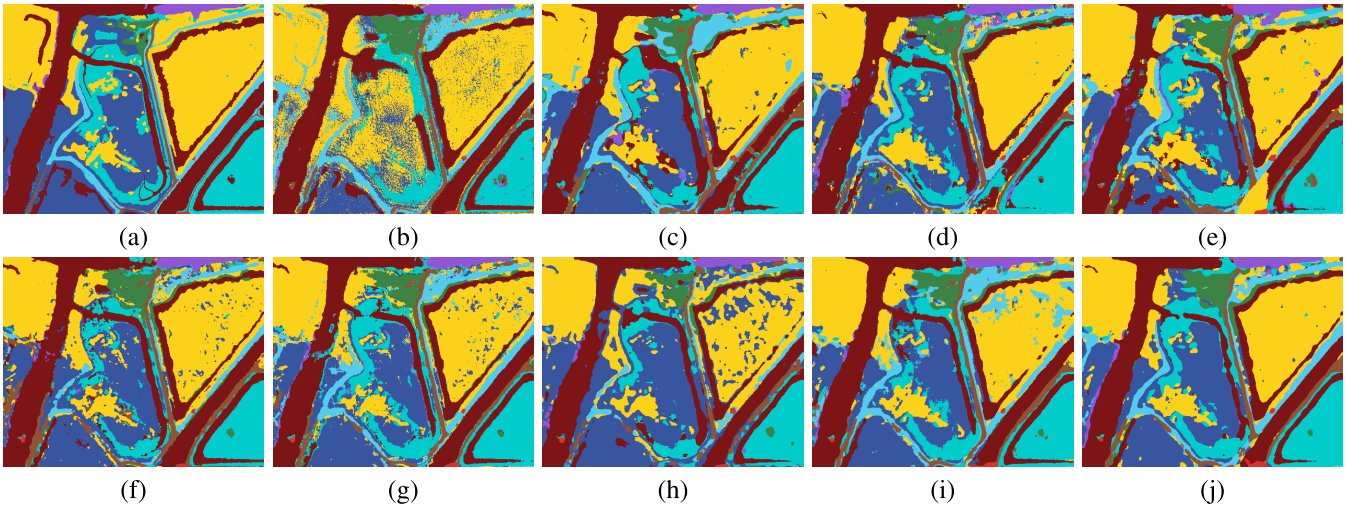


Fig. 12. Visual comparisons on Yancheng dataset. (a) Ground truth. (b) MSA-GCN(67.35%). (c) MICF-Net(76.6%). (d) M2FNet(75.85%). (e) ExViT(72.89%). (f) DSHFNet(76.81%). (g) MHST(77.2%). (h) MFT(74.01%). (i) MS2CANet(75.2%). (j) GAS^2-FL(80.44%). All maps share the same class-to-color mapping as the Yancheng ground-truth map in Fig. 7.

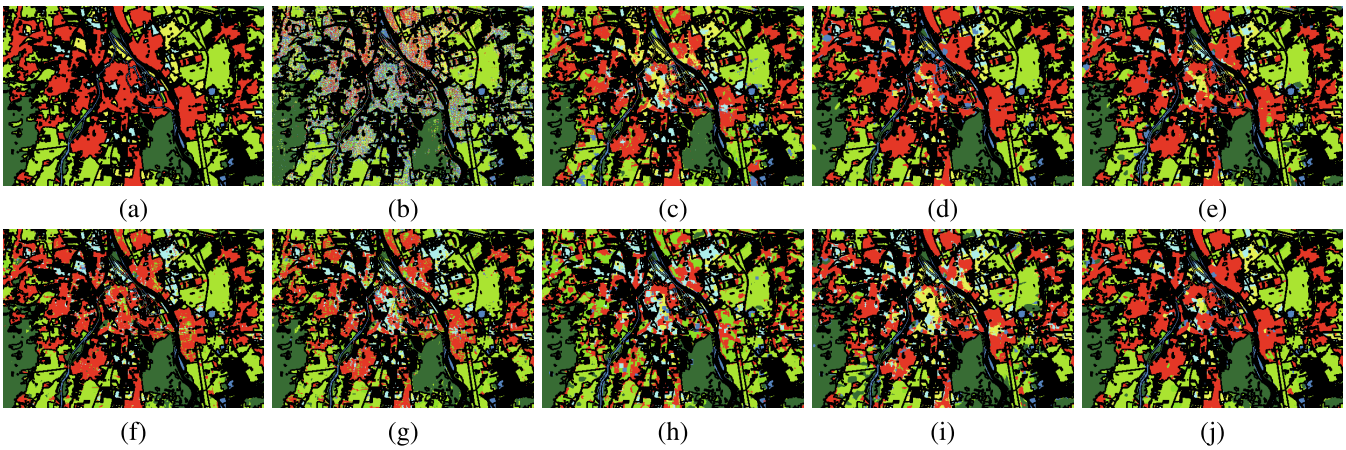


Fig. 13. Visual comparisons on Augsburg dataset. (a) Ground truth. (b) MSA-GCN(53.15%). (c) MICF-Net(86.85%). (d) M2FNet(86.68%). (e) ExViT(84.2%). (f) DSHFNet(84.56%). (g) MHST(76.23%). (h) MFT(62.39%). (i) MS2CANet(73.24%). (j) GAS^2-FL(91.35%). All maps share the same class-to-color mapping as the Augsburg ground-truth map in Fig. 8.

fragmented landscapes—demonstrated by the highest Kappa consistency and robust AA scores across all datasets. Notably, the framework maintains constrained accuracy deviations, thereby validating its architectural robustness across diverse hyperspectral imaging and LiDAR conditions.

To better demonstrate our classification accuracy, we provide visual comparisons of classification maps between our GAS^2-FL and related models across all four datasets (shown in Figs. 10–13). The visual results further validate the effectiveness of our GAS^2-FL in accurately classifying complex land covers.

To further support the class-wise behavior, we provide the row-normalized confusion matrices (class indices follow the same order as the CA rows in Tables I–IV) and the corresponding test-set error maps in Fig. 14.

For the Houston dataset (Fig. 10), our model achieves pixel segmentation, including Healthy Grass, Soil, Water, Highway, Tennis Court, and Running Track. The classification maps exhibit sharp boundaries between urban structures. For Yancheng dataset (Fig. 12), critical categories like Suaeda glauca and S. alterniflora are precisely delineated. For MUUFL dataset (Fig. 11), our model achieves exceptional accuracy in geometrically complex categories: Building Shadow, Sidewalk, and Yellow Curb. Furthermore, GAS^2-FL excels in natural landscapes, achieving 97.39% accuracy for Forest and Low Plants in the Augsburg dataset (Fig. 13). Although minor misclassifications are observed in the Industrial Area and Commercial Area, the model nonetheless preserves overall coherence within complex urban background.

D. Parameters Analysis

1) Effect of the Number of Groups in GSpectral-IFL and GSpatialFL: We evaluated how the number of groups N_g affects classification performance by testing values from 2 to 16 on four datasets. As shown in Fig 15, Houston2013, Yancheng, and Augsburg datasets achieve optimal OA at four groups, with performance declining while exceeding this

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

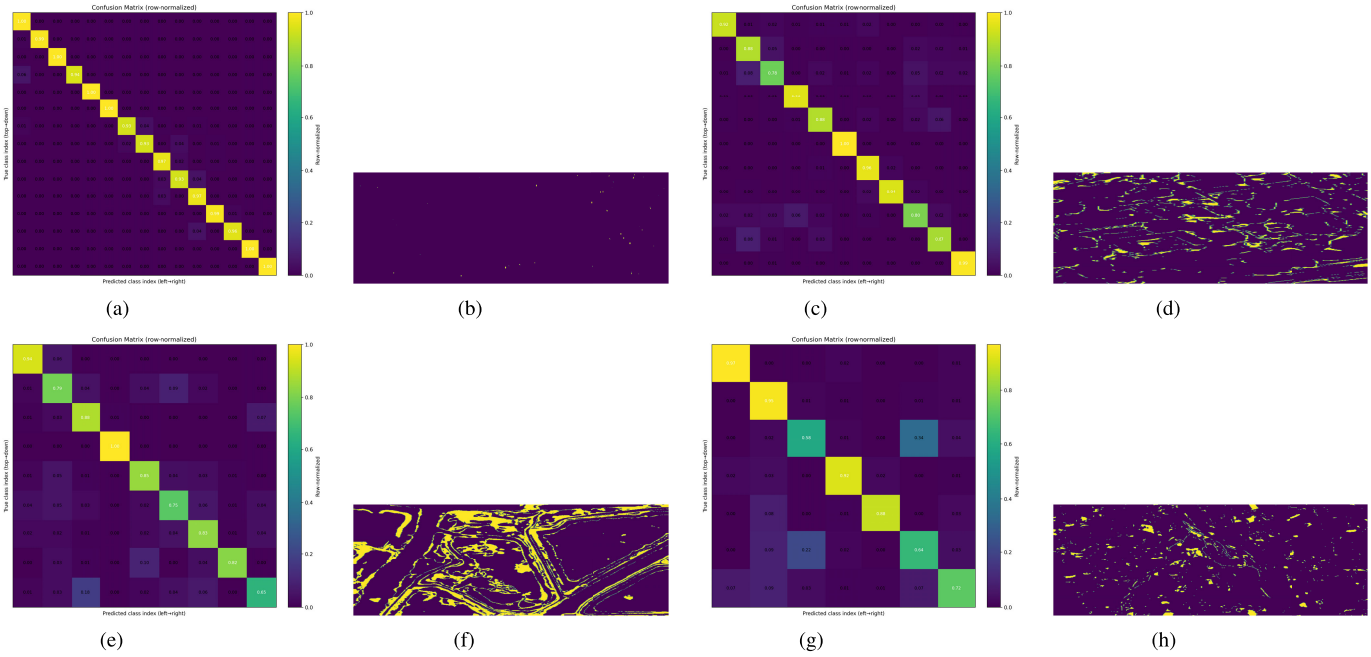


Fig. 14. Confusion matrices (row-normalized) and binary error maps on the test set. (a) Houston: confusion matrix. (b) Houston: error map. (c) MUUFL: confusion matrix. (d) MUUFL: error map. (e) Yancheng: confusion matrix. (f) Yancheng: error map. (g) Augsburg: confusion matrix. (h) Augsburg: error map. In each error map, correctly classified pixels are shown in dark and misclassified pixels are highlighted in yellow. The class index order follows the CA table order in Tables I–IV.

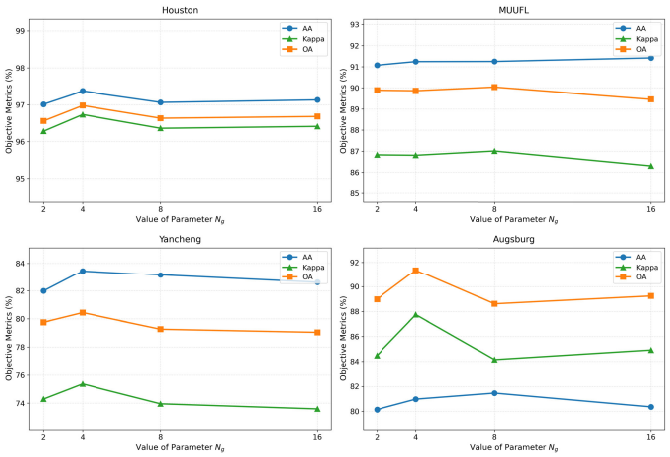


Fig. 15. Effect of the number of groups N_g in GSpectral-IFL and GSpatialFL on four datasets.

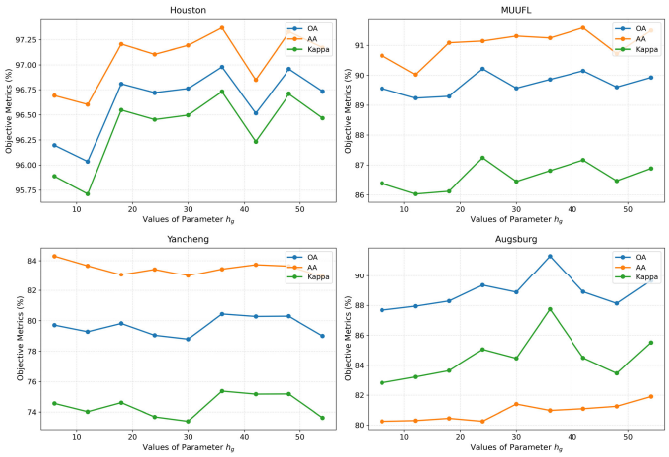


Fig. 16. Effect of different numbers of heads h_g in GSpectral-IFL and GSpatialFL.

threshold due to excessive feature fragmentation. It suggests that finer spatial grouping in the module may inadvertently fragment low-rank spatial priors, slightly compromising class separability. For the MUUFL dataset, all metrics maintains near-saturation levels. At $N_g = 16$, OA and Kappa slightly decrease for too many groups. We set $N_g = 4$ in our GAS^2-FL, as this configuration achieves the optimal balance across all datasets, preserving the module's low-rank prior.

2) Effect of the Number of Heads in GSpectral-IFL and GSpatialFL: We assessed the influence of the head number on classification performance by sweeping this value across all four datasets. We set the number of heads from 6 to 54. As shown in Fig. 16, the head number h_g, which controls the diversity of spatial and spectral attention patterns in the multi-head GSpectral-IFL and GSpatialFL modules, demonstrates a nonlinear relationship with feature fusion. For the Houston and Augsburg datasets, OA and Kappa peak sharply at $h_g = 36$, benefiting from the convolutional fusion layer's capacity to aggregate spatially refined attention maps from 36 heads. Redundant heads marginally degrade Kappa due to over-localized feature competition, while insufficient heads would limit spectral–spatial diversity in GSpatialFL and GSpectral-IFL. The Yancheng dataset exhibits lower sensitivity to head numbers, with h_g resolving the tradeoff between spectral and spatial attention. $h_g = 36$ is also a preferable choice for the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

TABLE V
EXPERIMENTAL RESULTS OF DIFFERENT NUMBERS OF GROUPS AND HEADS IN GSPATIALFL AND GSPECTRAL-IFL

Parameters		Houston			MUUFL			Yancheng			Augsburg		
groups	heads	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
2	72	96.86	97.28	96.61	89.27	90.71	86.05	79.19	83.25	73.90	88.26	80.72	83.62
4	36	**96.98**	**97.37**	**96.73**	**89.86**	**91.25**	**86.81**	**80.44**	83.44	**75.37**	**91.35**	**80.97**	**87.75**
8	18	96.54	97.02	96.26	89.20	90.37	85.99	79.11	**83.66**	73.82	88.06	80.93	83.42
16	9	96.07	96.56	95.75	89.04	90.61	85.80	79.72	83.42	74.41	87.77	79.57	82.87

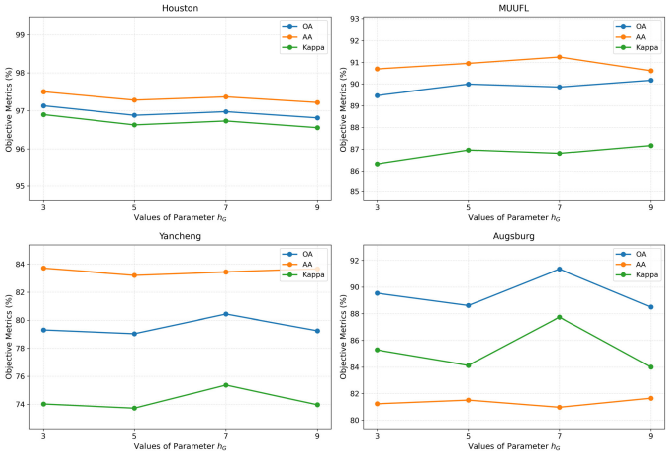


Fig. 17. Effect of different numbers of heads h_G in GPTransformer on four dataset.

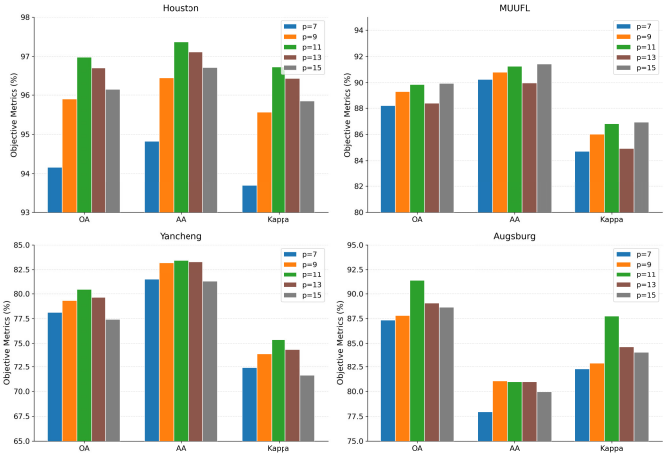


Fig. 18. Effect of different input patch size p on the four datasets.

MUUFL dataset, as it achieves the most favorable balance across all datasets.

Additionally, considering that the number of heads in GSpectral-IFL also impacts the number of groups, we set up several control groups that simultaneously select the number of groups and heads to explore more parameter possibilities. Table V represents the corresponding experimental results. We ensured that the final number of individual groups remained approximately equal, experimenting with various combinations of head and group counts. Comparative analysis revealed that the configuration with 36 heads and 4 groups consistently achieved the best performance on the Houston, MUUFL, and Augsburg datasets, and achieved near-optimal results on the Yancheng dataset. This indicates that these two parameters not only effectively balance the interaction between heads and groups, but also enhance the model's overall representational capability.

3) Effect of the Number of Heads in GPTransformer: There are two parameters for GPTransformer: the number of GPTransformer heads h_G and the number of groups. The head number in GPTransformer critically balances local–global feature fusion efficiency and multiscale land-cover discriminability. The constraints imposed by patches with small sizes limit the feasible selection of group numbers. For the number of GPTransformer heads, multiple parameter sets were evaluated, and the corresponding experimental results are presented in Fig. 17. For the Houston and MUUFL datasets, OA remains consistently high across h_G = 3–9, demonstrating GPTransformer's robustness on Houston dataset where grouped attention effectively captures repetitive spectral–spatial patterns regardless of head count. On the Yancheng and Augsburg datasets, $h_G = 7$ achieves optimal performance. However, performance declines at h_G = 9, suggesting excessive heads introduce redundant attention pathways. We select h_G = 7 as an experimental setting for the comprehensive performance across all datasets.

4) Effect of the Input Patch Size: The input patch size p is a critical parameter, which involves a tradeoff between local detail preservation and global context modeling. We set the patch size from 5 to 15 and conduct experiments on the four datasets. It is noted that as the input patch size changes we also change the downsampling strategy. While the input size cannot be divided by GPTransformer, our downsampling strategy before GPTransformer is to utilize a convolution with a kernel size of 3 with zero padding for downsampling. As the input size can be divided by GPTransformer, we utilize a convolution with padding and do not perform downsampling to control variables. The experimental results are shown in Fig. 18. The model's prediction accuracy is the highest as the patch size is 11 on the Houston, Yancheng, and Augsburg datasets. This patch size mitigates zero-padding distortions during downsampling convolution, whereas larger patches slightly impair performance due to spatial homogenization of low-rank features, and smaller patches restrict the spectral interaction scope within GSpectral-IFL. Although the prediction accuracy on the MUUFL dataset is not the highest, the discrepancy from the peak accuracy is minimal. Consequently,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

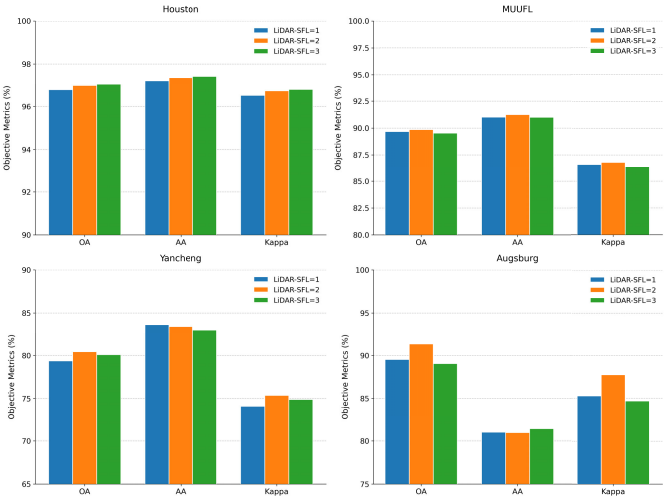


Fig. 19. Effect of different numbers of LiDAR_SFL on four datasets.

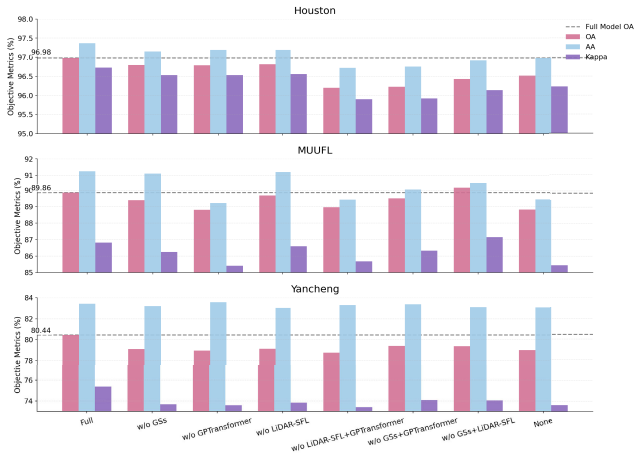


Fig. 20. Ablation study on LiDAR-SFL, GSpatialFL, GSpectral-IFL (GSs), and GPTransformer. "w/o" indicates removal of components: LiDAR-SFL, GSpatialFL, GSpectral-IFL, and GPTransformer. The "None" setting refers to the complete absence of all components, serving as the baseline.

considering all four datasets comprehensively, we adopt a patch size of 11 in our GAS^2-FL.

5) Effect of the Number of the Layers in LiDAR-SFL: We evaluated the impact of the layer number, a critical parameter in the LiDAR-SFL module, by sweeping it from 1 to 3 across all four datasets. The experimental results are shown in Fig. 19. As the number of layers of LiDAR-SFL is set to 2, the performance is the best. OA slightly increases from 96.78% (LiDAR-SFL = 1) to 96.98% (LiDAR-SFL = 2) and 97.04% (LiDAR-SFL = 3) on Houston dataset. For MUUFL, Yancheng, and Augsburg, the highest OA values are achieved at LiDAR-SFL = 2. To balance performance across all datasets and mitigate overfitting/underfitting risks, we set LiDAR-SFL = 2 as the optimal configuration.

E. Ablation Studies

Our network includes three branches: HSI branch, LiDAR branch, and a fused HSI-LiDAR branch. We conducted ablation studies to validate the spatial and spectral feature extraction capabilities of the GSpatialFL and GSpectral-IFL modules in HSI branch. Utilizing three objective evaluation metrics—OA, AA, and Kappa coefficient—we test these modules as shown in Fig. 20. Simultaneous activation of GSpatialFL and GSpectral-IFL with LiDAR-SFL and GPTransformer achieved a peak OA of 96.98% on Houston dataset, outperforming configurations with individual modules disabled. Specifically, disabling GSpatialFL reduced OA by 0.54%, while disabling GSpectral-IFL resulted in a 0.42% reduction. The complete deactivation of both GSpatialFL and GSpectral-IFL yielded an OA of 96.77%, demonstrating the necessity of their joint operation. The Yancheng dataset exhibited the most significant synergy—joint GSpatialFL and GSpectral-IFL activation improved OA by 1.49% over the baseline, surpassing the gains from individual module activations (0.93% for GSpatialFL and 1.02% for GSpectral-IFL). These results highlight the complementary roles of the modules in spectral–spatial feature extraction. While GSpatialFL alone attained superior AA on the Yancheng dataset, the integrated framework effectively balanced per-class discriminability with global consistency, as demonstrated by concurrent improvements in OA and Kappa metrics across all datasets.

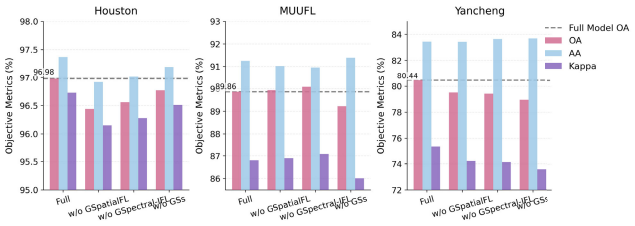


Fig. 21. Ablation study on GSpatialFL and GSpectral-IFL. "w/o" indicates removal of components: GSpatialFL and GSpectral-IFL. The "None" setting refers to the complete absence of all components, serving as the baseline.

Furthermore, the ablation studies validate the critical role of grouped architectural designs across all modules as shown in Fig. 21. Replacing the GSpectral-IFL and GSpatialFL modules with nongrouped operations, such as CBAM [32], resulted in a notable decline in performance, particularly on the Yancheng dataset, where OA decreased by 1.34% from 80.44% to 79.10%. These results show the critical role of these modules in addressing spectral and spatial ambiguities through low-rank constrained group interactions. Disabling the LiDAR-SFL module resulted in a decrease of 1.32% in OA and 1.51% in Kappa on the Yancheng dataset, confirming its pivotal role in enhancing optical spatial consistency. The GPTransformer module proved essential for effective cross-modal fusion; replacing its grouped token mechanism with standard Transformer layers caused significant OA reductions of 1.48% on Yancheng and 1.05% on MUUFL, underscoring the superiority of local–global attention in modeling heterogeneous sensor relationships. Notably, although the GPTransformer-only configuration unexpectedly improved OA on MUUFL by 0.29%, this is accompanied by declines in AA and performance across other datasets, indicating that ungrouped spatial and spectral processing produce feature noise despite OA improvements.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

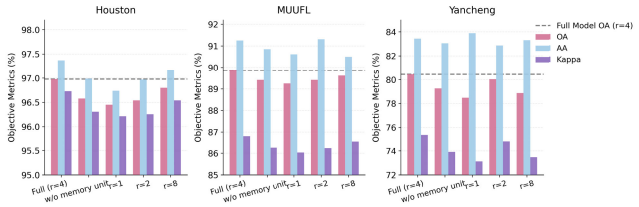


Fig. 22. Ablation of the rank-bounded memory unit under different subspace capacities on three datasets. The memory subspace dimension is set as $K = C_{Ng}/r$, where C_{Ng} is the channel dimension of each spectral group and r controls the strength of the low-rank bottleneck.

The experimental results clearly demonstrate the influence between the LiDAR-SFL module and GPTransformer's local–global modeling approach. When GSpatialFL and LiDAR-SFL operate independently with GPTransformer disabled, MUUFL's AA drops 2.03% despite maintaining 96.79% OA on Houston dataset, revealing that LiDAR-SFL's spatial attention alone cannot preserve fine-grained structural relationships. Conversely, standalone GPTransformer usage achieves peak MUUFL OA but suffers a 0.74% degradation in AA, indicating its global modeling requires LiDAR-SFL's or GSpatialFL's localized spatial refinement. Furthermore, the Yancheng dataset shows the best module gain—simultaneous activation of all grouped components yielded 1.43% OA and 1.74% Kappa improvements over the baseline.

We further study how the memory unit behaves under different subspace capacities by sweeping the rank ratio r, which determines the latent memory dimension as $K = C_{Ng}/r$, where C_{Ng} denotes the channel dimension of each spectral group. A smaller r yields a larger subspace, and $r = 1$ corresponds to $K = C_{Ng}$ without an explicit rank bottleneck, while a larger r imposes a stronger low-rank constraint by reducing K.

Fig. 22 summarizes OA/AA/Kappa on Houston2013, MUUFL, and Yancheng. Removing the memory unit leads to consistent degradation across all datasets. With $r = 4$, the full model achieves 96.98% OA and 96.73% Kappa on Houston2013, while the variant without memory drops to 96.58% OA and 96.31% Kappa. Similar decreases are observed on MUUFL, where OA changes from 89.86% to 89.42%, and on Yancheng, where OA decreases from 80.44% to 79.26%. These results confirm that the memory unit contributes nontrivially even when the backbone remains unchanged.

Varying r further indicates that a moderate bottleneck offers the most reliable gains. Compared with $r = 1$, which provides the largest subspace but no explicit rank compression, setting $r = 4$ improves OA by 0.53% on Houston2013 and by 1.96% on Yancheng, suggesting that the bottleneck helps suppress redundant channel responses while preserving class-discriminative cues. When the constraint is too strong, e.g., $r = 8$ with a very small K, performance saturates or slightly decreases on Yancheng. On MUUFL, $r = 2$ yields a marginally higher AA than $r = 4$, yet $r = 4$ maintains the best OA with competitive Kappa. Therefore, we adopt $r = 4$ as the default configuration in the remaining experiments.

In addition, we conduct a significance test on the Houston dataset in terms of OA to verify that the Full model consistently outperforms the ablation variants. Specifically, we compare the OA of the Full model against each ablation counterpart over five runs using the same set of random seeds, and perform a paired two-sided Student's t-test. The Full model achieves significantly higher OA than all ablation variants on Houston (significant at the 0.05 level).

V. Conclusion

In this article, we propose a novel GAS^2-FL network for hyperspectral and LiDAR collaborative classification. The innovation lies in three key components: The GSpectral-IFL module explicitly models spectral correlations through groupwise low-rank constraints and cross-group attention, effectively preserving discriminative spectral patterns while suppressing redundant information. The GSpatialFL module adopts adaptive spatial attention mechanisms within spectral groups, capturing hierarchical spatial structures and elevation characteristics from LiDAR data. The GPTransformer integrates nonlocal learning and groupwise self-attention, achieving balanced local–global feature fusion through dynamic interaction between grouped tokens and global context. Comprehensive testing across four standard datasets reveals the GAS^2-FL method substantially surpasses competing approaches. Notably, it achieves OA improvements of 1.4% (Houston2013), 1.87% (MUUFL), 3.24% (Yancheng), and 4.5% (Augsburg) over the second-best methods. The framework shows strong robustness across different land covers.

Future work will focus on extending this architecture to multitemporal remote sensing data fusion and developing lightweight variants for edge computing applications. The groupwise feature enhancement paradigm demonstrates considerable potential for broader applications in multimodal interpretation tasks within the field of remote sensing. Meanwhile, we will prioritize reducing model size and computational cost by designing lightweight modules and efficient attention schemes, enabling faster inference and lower memory footprint without sacrificing accuracy.

References

[1] J.-Y. Yang, H.-C. Li, J.-H. Yang, L. Pan, Q. Du, and A. Plaza, "Multifrequency graph convolutional network with cross-modality mutual enhancement for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505914.

[2] W.-S. Hu, W. Li, H.-C. Li, F.-H. Huang, and R. Tao, "Global clue-guided cross-memory quaternion transformer network for multisource remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 7357–7371, Apr. 2025.

[3] J. Feng, J. Zhang, and Y. Zhang, "Multiview feature learning and multilevel information fusion for joint classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528613.

[4] S. Xia, X. Zhang, H. Meng, and L. Jiao, "Ternary modality contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522017.

[5] K. Ni, D. Wang, G. Zhao, Z. Zheng, and P. Wang, "Hyperspectral and LiDAR classification via frequency domain-based network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5525117.

[6] Y. Yang, J. Qu, W. Dong, T. Zhang, S. Xiao, and Y. Li, "TMCFN: Text-supervised multidimensional contrastive fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511015.

[7] S. Jia, X. Zhou, S. Jiang, and R. He, "Collaborative contrastive learning for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507714.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.

[8] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[9] T. Zhang, S. Xiao, W. Dong, J. Qu, and Y. Yang, "A mutual guidance attention-based multi-level fusion network for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[10] X. Shen, X. Deng, J. Geng, and W. Jiang, "SFE-FN: A shuffle feature enhancement-based fusion network for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[11] W. Dong, T. Zhang, J. Qu, S. Xiao, T. Zhang, and Y. Li, "Multibranch feature fusion network with self- and cross-guided attention for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530612.

[12] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506812.

[13] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.

[14] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-Net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

[15] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522514.

[16] Y. Zhao, W. Bao, J. Xu, and X. Xu, "BIHAF-Net: Bilateral interactive hierarchical adaptive fusion network for collaborative classification of hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 15971–15988, 2024.

[17] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[18] S. Zhang, X. Meng, Q. Liu, G. Yang, and W. Sun, "Feature-decision level collaborative fusion network for hyperspectral and LiDAR classification," *Remote Sens.*, vol. 15, no. 17, p. 4148, Aug. 2023.

[19] W. Dong, T. Yang, J. Qu, T. Zhang, S. Xiao, and Y. Li, "Joint contextual representation model-informed interpretable network with dictionary aligning for hyperspectral and LiDAR classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 6804–6818, Nov. 2023.

[20] Y. Pan, N. Wu, and W. Jin, "Multimodal feature disentangle-fusion network for hyperspectral and LiDAR data classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[21] H. Zhang, J. Yao, L. Ni, L. Gao, and M. Huang, "Multimodal attention-aware convolutional neural networks for classification of hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3635–3644, 2023.

[22] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530416.

[23] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[24] M. Wang, Y. Sun, J. Xiang, and Y. Zhong, "CITNet: Convolution interaction transformer network for hyperspectral and LiDAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535918.

[25] L. Sun, X. Wang, Y. Zheng, Z. Wu, and L. Fu, "Multiscale 3-D–2-D mixed CNN and lightweight attention-free transformer for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2100116.

[26] Y. Feng, J. Zhu, R. Song, and X. Wang, "S2EFT: Spectral–spatial-elevation fusion transformer for hyperspectral image and LiDAR classification," *Knowl.-Based Syst.*, vol. 283, Jan. 2024, Art. no. 111190.

[27] W. Wang et al., "Dual-branch feature fusion network based cross-modal enhanced CNN and transformer for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] X. Du, X. Zheng, X. Lu, and X. Wang, "Hyperspectral and LiDAR representation with spectral–spatial graph network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9231–9245, 2023.

[29] J. Li, Y. Liu, R. Song, Y. Li, K. Han, and Q. Du, "Sal^{2}RN: A spatial–spectral salient reinforcement network for hyperspectral and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500114.

[30] X. Wang, J. Zhu, Y. Feng, and L. Wang, "MS2CANet: Multiscale spatial–spectral cross-modal attention network for hyperspectral image and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[31] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[32] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[33] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1971–1980.

[34] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11534–11542.

[35] Y. Mei et al., "Pyramid attention network for image restoration," *Int. J. Comput. Vis.*, vol. 131, no. 12, pp. 3207–3225, Dec. 2023.

[36] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based SpectroSpatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 92–93.

[37] Z. Wang, Q. Wang, J. Zhang, and X. Liang, "Joint classification of hyperspectral and LiDAR data based on heterogeneous attention feature fusion network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 8616–8619.

[38] Z. Li, H. Sui, C. Luo, and F. Guo, "Morphological convolution and attention calibration network for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5728–5740, 2023.

[39] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2021.

[40] Y. He et al., "Multilevel attention dynamic-scale network for HSI and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5529916.

[41] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[42] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[43] Z. Gao, Q. Wang, B. Zhang, Q. Hu, and P. Li, "Temporal-attentive covariance pooling networks for video recognition," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 13587–13598.

[44] M. Li, J. Liu, Y. Fu, Y. Zhang, and D. Dou, "Spectral enhanced rectangle transformer for hyperspectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Apr. 2023, pp. 5805–5814.

[45] Z. Huang et al., "Feature shrinkage pyramid for camouflaged object detection with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5557–5566.

[46] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[47] H. Gao et al., "Interactive enhanced network based on multihead self-attention and graph convolution for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5533716.

[48] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[49] K. Ni, D. Wang, Z. Zheng, and P. Wang, "MHST: Multiscale head selection transformer for hyperspectral and LiDAR classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5470–5483, 2024.

[50] X. Tang, Y. Zou, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Multiple information collaborative fusion network for joint classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5525416.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.



Yan Huang received the B.S. degree in geophysics from Jilin University, Changchun, China, in 2008, and the M.S. degree in geological engineering from Hehai University, Nanjing, China, in 2020. He is currently pursuing the Ph.D. degree in computer science with Nanjing University of Science and Technology (NJUST), Nanjing.

He is the Director of the Airborne Geophysics and Remote Sensing Department, Geological Exploration Technology Institute of Jiangsu Province, Nanjing. His research interests include airborne geophysics and remote sensing.



Binghua Lu is currently pursuing the bachelor's degree in software engineering with Nanjing University of Posts and Telecommunications, Nanjing, China.

His research interests include multi-source remote sensing image processing.



Zhizhong Zheng received the B.S. degree in electronic engineering from Jilin University, Changchun, China, in 2001, and the Ph.D. degree in control engineering from Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China, in 2021.

He is currently the Team Leader of hyperspectral remote sensing and intelligent perception with Nanjing University of Posts and Telecommunications (NJUPT), Nanjing. He has developed a series of equipments such as space remote sensing camera, airbone hyperspectral imaging instrument, UAV hyperspectral remote sensing systems, little field spectrometer, and core scan spectral imaging systems, as well as spectral data processing and information extraction software.



Kang Ni (Member, IEEE) received the M.S. degree from Changchun University of Technology, Jilin, China, in 2016, and the Ph.D. degree from Nanjing University of Aeronautics and Astronautics, Nanjing, Jiangsu, China, in 2020.

He is an Associate Professor with the School of Computer Science, Nanjing University of Posts and Telecommunications, Nanjing, and also a member with Jiangsu Key Laboratory of Big Data Security and Intelligent Processing, Nanjing. His research interests include machine learning, SAR image processing, and multisource remote sensing image processing.



Liang Xiao (Senior Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 1999 and 2004, respectively.

From 2009 to 2010, he was a Post-Doctoral Fellow with Rensselaer Polytechnic Institute, Troy, NY, USA. Since 2014, he has been the Deputy Director of Jiangsu Key Laboratory of Spectral Imaging Intelligent Perception, Nanjing. He was the Second Director of the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information, Ministry of Education, NJUST, where he is currently a Professor with the School of Computer Science. He has authored or co-authored more than 70 international journal articles, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. His main research interests include inverse problems in image processing, computer vision and image understanding, pattern recognition, and remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:00 UTC from IEEE Xplore. Restrictions apply.